Tamanjyot Singh

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Research Interests

Relativistic Astrophysics | Black Hole Spin | Tidal Disruption Events | Gravitational Waves (GWs)
Spin Precession in Binary Black Holes | Gravitational Lensing of GWs

Education

Ph.D. **The University of Texas at Dallas** — Richardson, TX, USA
Physics — January 2021 - March 2026
Thesis: *The Black Hole Ballet*: Spin and precession effects on gravitational waves and tidal disruption events
Advisor: Dr. Michael Kesden

M.S. **The University of Texas at Dallas** — Richardson, TX, USA
Physics — January 2021 - December 2023
GPA: 3.93/4.0

B.Tech. **Punjab Engineering College (Deemed to be University)** — Chandigarh, India
Aerospace Engineering with minor in Mathematics — August 2016 - June 2020
GPA: 9.2/10.0, Rank 2

Experience

Postdoctoral Fellow — Baltimore, MD, USA
William H. Miller III Department of Physics & Astronomy, Johns Hopkins University — September 2026 - Present
Advisor: Prof. Emanuele Berti

Research Assistant — Richardson, TX, USA
Department of Physics, The University of Texas at Dallas — May 2022 - August 2026
Advisor: Prof. Michael Kesden

Teaching Assistant — Richardson, TX, USA
Department of Physics, The University of Texas at Dallas — January 2021 - May 2022
Supervisor: Dr. Kuei Sun

Research Intern — Mumbai, India
Combustion Research Laboratory, Indian Institute of Technology, Bombay — January 2019 - April 2019
Advisor: Prof. Sudarshan Kumar

Indian Air Force (IAF) Intern — Chandigarh, India
3rd Base Repair Depot (3BRD), Air Force Station — June 2018

Publications

Published:

T. Singh, M. Kesden — February 2024
"Distribution of orbital inclinations for tidal disruption events by Kerr black holes"
Phys. Rev. D 109, 043016, ADS

Submitted:

T. Singh, E. Stoikos, S. Ali, N. Steinle, M. Kesden, and L. King — September 2025
"Detecting regular precession using a new gravitational waveform model directly parameterized by both precession amplitude and frequency"
Submitted to *Phys. Rev. D*, arXiv: 2509.10628, ADS

T. N. Nguyen-Vo[†], ***T. Singh***[†], B. McCallip, M. Kesden, and L. King ([†] Co-leads) September 2026
"Distinguishing lensing and precessional modulation in binary black-hole inspiral waveforms"
Submitted to *Phys. Rev. D*, arXiv: 2609.06347, ADS

In preparation:

T. Singh, M. Kesden, and L. King
"Detecting nutation using `generic_precession` waveform model"

Research Presentations

Contributed	**APS Global Physics Summit 2026**, Denver, CO	March 2026
Contributed	**Fall meeting 2025, Texas, Arkansas, Oklahoma Section APS**, Texas Christian University, Fort Worth, TX	October 2025
Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2025**, The University of Texas at Dallas, Richardson, TX	April 2025
Contributed	**APS Global Physics Summit 2025**, Anaheim, CA	March 2025
Contributed	**Fall meeting 2024, Texas Section APS**, Southern Methodist University, Dallas, TX	October 2024
Invited Visit	Center for Computational Astrophysics, Flatiron Institute, New York City, NY	May 2024
Contributed	**APS April meeting 2024**, Sacramento, CA	April 2024
Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2024**, University of North Texas, Denton, TX	March 2024
Contributed	**10th Gulf Coast Gravity Meeting 2024**, University of Texas, Austin, TX	March 2024
Contributed	**International Conference on Gravitation and Cosmology 2023**, Indian Institute of Technology, Guwahati, Assam, India	December 2023
Contributed	**Fall meeting 2023, Texas Section APS**, Angelo State University, San Angelo, TX	October 2023
Contributed	**Spring meeting 2023, Texas Section APS**, Texas A&M University, Commerce, TX	March 2023

Awards and Recognitions

NSM Conference Travel Award March 2025
School of Natural Sciences and Mathematics, The University of Texas at Dallas, TX

APS April meeting DGRAV Travel Award April 2024
Division of Gravitational Physics, APS

Margie Renfrow Travel Award March 2024
Department of Physics, The University of Texas at Dallas, TX

Teaching Assistant Award 2022 December 2022
Department of Physics, The University of Texas at Dallas, TX

PECOSA Sydney Prize (Awarded to 3 out of 700+ students) September 2018
Punjab Engineering College, Chandigarh

1st position in the Mathematics Quiz September 2016
Punjab State Council for Science and Technology, Chandigarh

Mentoring

I have had the pleasure of mentoring with many undergraduate and near-peer mentees over the years:

Wanrou Sun	Summer REU student 2026 *(Undergrad student)*
Austin Crawford	UTD grad student Fall 2024 - Present *(Grad Student)*
Benjamin McCallip	Summer REU student 2025 *(Undergrad student)*
Tien Nguyen	PAARE fellow 2023-24 *(Post-baccalaureate student)*
Maya Basu	Summer REU student 2024 *(Undergrad Student)*
Tinatin Tsiskaridze	Summer exchange student 2024 *(Grad Student)* (from Tblisi State University, Georgia)
Niel Shah	Summer REU student 2023 *(Undergrad Student)*
Deven Linthicum	Summer REU student 2023 *(Undergrad Student)*
Allison Powell	Summer REU student 2022 *(Undergrad Student)*

Service and Leadership

Reviewer — Physical reviews D, Monthly Notices of the Royal Astronomical Society

Journal Club Host: Hosted weekly journal club meetings with astrophysics students and faculty across DFW in a joint effort between the University of Texas at Dallas, University of North Texas, Denton, and University of Texas at Arlington. — August 2024 - August 2025

Course Facilitator: Volunteered to teach and translate Punjabi along with hosting biweekly community sessions, fostering an environment for positive outlook on life as Sikhs in the USA. — August 2024 - May 2025

Graduate Students in Physics - Vice President: Representing graduate student interests in the department. Assisting President in maintenance and operations of the organization. *Notable*: Procured a department-wide Overleaf license. — May 2023 - July 2024

Graduate Students in Physics - Astronomy committee head: Graduate student liaison for all things telescope, hosting star-watch and lunar eclipse parties. — March 2022 - February 2024

Technical skills

I built a couple of web-apps for my GW projects: `Regular Precession` and `Generic Precession`.

- Python: `numpy`, `matplotlib`, `scipy`, `PyCBC`, `astropy`, etc. (Developed: `regular_precession`)
- Other programming languages: C, C++, Matlab
- General: `git`, LaTeX, `HTML`, cluster computing (`slurm`)

Professional Development

Participant in **Prospects in Theoretical Physics 2025** — July 2025
Institute for Advanced Study, Princeton, NJ

Participant in **Entering Mentoring** — July 2024
The University of Texas at Dallas, Richardson, TX

Course Certificate for **Astro 101 : Black Holes** — June 2020
University of Alberta on Coursera.

Host for workshop on **Gravitational lensing** (Undergraduate) — October 2019
Punjab Engineering College, Chandigarh

Participant in **ASTROWIN**, Winter School on Astronomy 2019 — February 2019
Birla Museum, Hyderabad

Course Certificate for **Particle Physics: an Introduction** — July 2018
University of Geneva on Coursera.

Course Certificate for **Understanding Einstein: The Special Theory of Relativity** — July 2017
Stanford University on Coursera.